

Xiwen Huang

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Education

Southern University of Science and Technology (SUSTech)

Shenzhen, China

Bachelor of Engineering in Information Engineering

Expected Jun.2027

GPA: 3.76/4.0, Honor Class

Analog Circuits Laboratory (100/100), Digital Circuits Laboratory (99/100), Fundamentals of Electric Circuits (95/100), Digital Signal Processing (95/100), Digital Circuits (92/100), Signals and Systems (92/100), Engineering Electromagnetics (90/100), Data communications and networking (90/100), Introduction to Computer Programming (96/100), Calculus II (94/100)

University of Pennsylvania

Philadelphia, USA

Exchange Program

Jan. 2026 - May 2026

GPA: 4.0 / 4.0

Machine Learning (A, 4.0/4.0), Foundations of AI: Deep Learning with Applications (A, 4.0/4.0), Musical Interfaces and Robotics (A+, 4.0/4.0)

Research Interest

I study safe and scalable robot learning for real-world deployment. My research interests center on reinforcement learning, sim-to-real transfer, and embodied agents that can perceive, reason, and act robustly in complex environments. I aim to combine principled learning methods with practical system design to advance reliable robotic systems that generalize beyond controlled environments.

Honors & Awards

2026, 2024: SUSTech Study Abroad Scholarship

2025: Honorable Mention, Mathematical Contest in Modeling (MCM) — Team Leader

2024: SUSTech, Department of Electronics and Electrical Engineering - Wo Long Elite Class (Top 8%, Honor Class)

2024: Sole Winning Team, Nvidia Program of Blended Learning, MIT xPRO

2024: SUSTech Outstanding Student

2024: SUSTech Excellence Scholarship

Publication

CA-LRR: Controller-Aware Low-Rank Residual Refinement for Contact-Rich Aerial Manipulation

Second author; submitted to the IEEE International Conference on Robotics and Automation (ICRA 2027).

Research & Project Experience

CA-LRR for Contact-Rich Aerial Manipulation

Research Assistant; Westlake University; Advisor: Shiyu Zhao

Jun. 2026 - Sep. 2026

- Contributed to implementing a low-rank residual RL framework for coordinated base-arm refinement around a frozen reference policy, using a five-dimensional SVD basis derived from successful simulation rollouts.
- Implemented controller-aware residual scaling based on tracking errors relative to prescribed bounds, with direction-preserving scaling to enforce action and joint limits.
- Engineered ROS-based hardware deployment integrating Vicon pose feedback, PreGME flight control, and reference and residual policy inference at 8 Hz.
- The resulting method increased mean task success from 44.91% to 56.72% across nine simulated load-slope conditions. Hardware experiments demonstrated pushing a cart with a total mass of 35.54 kg and, in a separate test, pushing across a 2 cm-high uneven surface.

LLM Hallucination Research

Research Assistant; University of Pennsylvania; Advisor: Enric Boix

Feb. 2026 - May 2026

- Fine-tuned TinyLlama-1.1B on SQuAD 2.0 using PEFT, introducing a custom abstention objective that enabled the model to explicitly abstain when uncertain.
- Reduced incorrect non-abstaining responses by 12% on a held-out test split compared with the baseline model.

Universal Vessel-like Structure Segmentation

Research Assistant; SUSTech; Advisor: Xiaoying Tang Jul. 2025 - Dec. 2025

- Developed a universal vessel-like structure segmentation model based on nnU-Net incorporating prompt fusion and dual attention, achieving the segmentation of 10+ vessel-like structures across multiple modalities in a single network.
- Improved average Dice from 84.92 to 85.79 (+0.87) and average cIDice from 89.58 to 91.38 (+1.80) on 7 2D datasets, outperforming single-task nnU-Net baselines.

Learning Quadrupedal Robot Locomotion Policy with APG in Differentiable Simulation

Research Assistant; SUSTech; Advisor: Jiankun Wang Feb.2025 - Jun. 2025

- Implemented differentiable-physics-based policy learning with MuJoCo XLA (MJX), empirically demonstrating that this approach achieved better sample efficiency than PPO by training quadruped simulations and residual motion policies via analytical policy gradients.
- Adapted the ANYmal simulation environment to the Unitree AI platform; built an interactive test visualization tool and implemented parameter saving/loading to ensure experiment reproducibility and fast deployment.

Professional Experience

Shennan Circuits Co., Ltd Shenzhen, China

Intern, IT Process and Control Department Jan. 2025 - Feb. 2025

- Led full-stack development of a shuttle bus waiting system with an HTML/CSS front-end and Java back-end, enabling 50+ employees to check real-time bus arrival times.
- Designed and managed an SQL database for bus routes, schedules, and real-time passenger waiting data.

Skill Set

Programming: Python, C, C++, Java, MATLAB, HTML/CSS, SQL, LaTeX, LabVIEW

Frameworks & Tools: PyTorch, PEFT, MuJoCo, ROS, PX4, Gazebo

Hardware & Embedded: Raspberry Pi, Embedded Systems, Circuit Design

Languages

Mandarin Chinese: Native

English: TOEFL iBT 5.5

Japanese: Elementary

Spanish: Elementary